Associations between use of targeted therapies, hospitalizations, and emergency department visits among patients with atopic dermatitis: A retrospective cohort study

Richard Ta, PharmD, MS; Jong Min Jung, MS, MA; Kyu Eun Lee, PhD; Jing Li, PhD

Plain language summary

Atopic dermatitis (AD) is a common long-term skin disease that can be severe for some adults. We compared adults who started new drugs for AD with adults who did not use them. Adults who started these drugs had fewer hospital stays and fewer emergency department (ED) visits. Their total health care costs were higher because the drugs were expensive. However, patients paid only slightly more out of pocket.

Implications for managed care pharmacy

This study shows that targeted AD therapies reduced hospitalizations and ED visits despite higher drug costs. These findings support benefit designs that prioritize access to effective biologics and Janus kinase inhibitors to offset costly acute care. Results may inform value-based coverage, step therapy refinement, and formulary placement favoring high-value agents. Improving access to targeted therapies may also reduce inequities by limiting preventable hospital-based care among vulnerable populations.

Author affiliations

The CHOICE Institute, Department of Pharmacy, University of Washington, Seattle (Ta, Lee, Li); Department of Economics, University of Washington, Seattle (Jung).

AUTHOR CORRESPONDENCE:
Jing Li, +1.206.543.6788;
jli0321@uw.edu

J Manag Care Spec Pharm.
2026;32(11):1377-385

Copyright © 2026, Academy of Managed Care Pharmacy. All rights reserved.

ABSTRACT

BACKGROUND: Atopic dermatitis (AD) is the most common inflammatory dermatologic condition. Despite approvals of numerous targeted therapies, no studies have examined their real-world relationship with acute health care uses and health care spending for patients with AD.

OBJECTIVE: To assess the association between use of targeted therapy and the probability of hospitalizations and emergency department (ED) visits and health care spending among adult patients with AD in the United States.

METHODS: This retrospective cohort study used MarketScan commercial insurance claims data from January 1, 2012, through December 31, 2023, with analyses conducted between November 1, 2023, and August 1, 2024. Adult patients (aged ≥18 years) with a diagnosis of AD who had at least 1 prescription claim for a targeted AD therapy (upadacitinib, abrocitinib, tralokinumab, baricitinib, dupilumab) were included. The primary outcomes were any hospitalizations and any ED visits in a given year. The secondary outcomes were annual total health care spending and spending by type of service (drug, inpatient, and outpatient), and patient out-of-pocket costs. A difference-in-differences (DiD) design was employed to compare changes in outcomes between patients with AD who received targeted therapies and those who did not receive a prescription for these therapies.

RESULTS: Among 157,966 (mean [SD] age = 41.6 [13] years; 55,077 [35%] male) individuals with AD, 9,320 (6%) received prescriptions for a targeted therapy, among whom 99% received dupilumab. The DiD results showed that targeted-therapy use was associated with a decrease of 1.6 percentage points (95% CI = −0.021 to −0.010; $P < 0.01$) in the probability of any hospitalization and a decrease of 1.7 percentage points (95% CI = −0.025 to −0.008; $P < 0.01$) in the probability of any ED visits. Targeted therapy was associated with an increase in overall health care spending ($24,901; 95% CI = $24,482-$25,320), which was entirely due to increases in drug spending ($26,061; 95% CI = $25,735-$26,388) along with decreases in inpatient

ABSTRACT *continued*

and outpatient spending. It was also associated with a $95 increase in annual patient out-of-pocket spending (95% CI = $83-$108).

CONCLUSIONS: Among adults with AD, initiation of targeted therapy was associated with fewer all-cause hospitalizations and ED visits relative to nontargeted-therapy care. Targeted therapy was associated with higher total and drug spending but lower inpatient and outpatient spending.

Atopic dermatitis (AD) is the most prevalent inflammatory dermatologic disorder, affecting approximately 2% to 10% of adults and 10% to 30% of children.[1] Its common manifestations include pruritus, dry skin, rashes, and erythema.[1] If left untreated, AD significantly impacts patients' quality of life, leading to sleep disturbances, decreased productivity at school or work, and adverse effects on mental health.[2] The chronic nature and increasing prevalence of AD contribute to a substantial economic burden in the United States. In 2015, the estimated annual cost of AD in the United States amounted to approximately $5.3 billion.[3] Furthermore, a study revealed a notable rise in AD-related total emergency department (ED) costs from $265 million in 2006 to $642 million in 2012.[4]

Conventional immunosuppressants were the main systemic treatment for moderate-to-severe AD until 2017.[5] Since then, the treatment landscape has changed substantially with the approval of systemic targeted therapies, including biologics such as dupilumab and tralokinumab as well as oral Janus kinase (JAK) inhibitors such as abrocitinib, upadacitinib, and baricitinib.[6] These therapies target inflammatory pathways involved in AD and have been shown to significantly improve disease severity, skin clearance, and quality of life.[7,8]

Hospitalizations and ED visits are critical outcomes for patients, clinicians, and payers, as they represent high-intensity and high-cost health care use. Adults who are diagnosed with AD may need these visits when their condition worsens or becomes difficult to manage, including sudden flare-ups, skin infections, complications from other health problems, or when their current treatment is no longer effective or causes side effects. By improving disease control and alleviating symptoms, these targeted therapies may significantly reduce the need for acute and urgent care. However, evidence supporting these therapies has relied primarily on clinical severity and symptom endpoints from randomized trials, leaving very limited real-world evidence, which is largely descriptive in nature, on whether these therapies are associated with lower hospitalizations, ED visits, and health care spending relative to nontargeted-therapy care.[9,10]

In this study, we address this gap by examining the association between targeted-therapy initiation and all-cause hospitalization, all-cause ED visits, and health care spending among commercially insured adults diagnosed with AD. We used a difference-in-differences (DiD) design to compare changes in these outcomes before and after the index date between individuals who received targeted therapy and those who did not.

Methods

DATA AND STUDY PERIOD

This retrospective cohort study used data from the Merative MarketScan commercial database, specifically the Commercial Claims and Encounters database, from January 1, 2012, through December 31, 2023.[11] The study received exempt status by the institutional review board of the University of Washington. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology reporting guidelines.

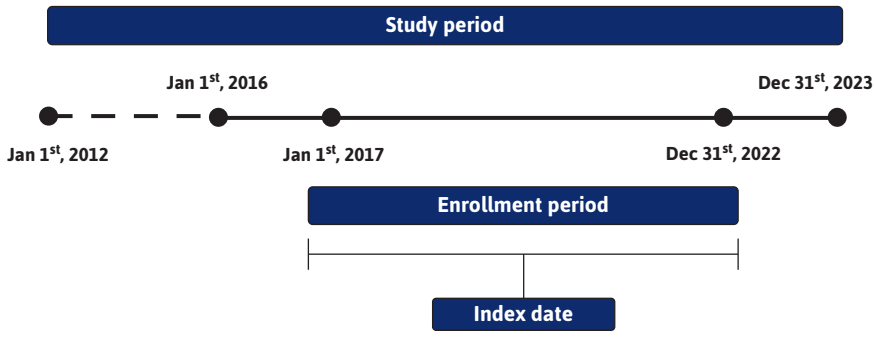
STUDY POPULATION AND EXPOSURE

We identified adults with AD during the patient enrollment period from January 1, 2017, through December 31, 2022 (Figure 1). AD was defined as having at least 1 inpatient claim or at least 2 outpatient claims with the *International Classification of Diseases, Tenth Revision* (ICD-10) diagnosis codes beginning with L20 in any billing position (Supplementary Table 1, available in online article), consistent with prior claims-based studies of adult AD.[12,13] Patients were included if they were aged 18 years or older and continuously enrolled in medical and pharmacy benefits for at least 12 months immediately before and after the index date (defined below).

The targeted-therapy group included adults with AD who received abrocitinib, dupilumab, tralokinumab, baricitinib, and upadacitinib during the patient enrollment period.[14] Targeted therapies were identified using National Drug Codes (Supplementary Table 2), and the index date was the date of the first observed targeted-therapy prescription claim. To identify new users of targeted therapy, patients were required to have no targeted-therapy prescription claim during the 12-month baseline washout period before the index date, consistent with new-user designs commonly used in pharmacoepidemiologic claims studies.[15]

The comparison group included adults with AD who never received any targeted therapy during the study period (January 1, 2012-December 31, 2023) and were continuously enrolled in medical and pharmacy benefits for at least 24 months between January 1, 2016, and December 31, 2023. Because these patients may still have received topical

FIGURE 1 Study Design



Study period: duration enrolled individuals may be analyzed, minimum of 24 month period

Index date (within enrollment period):

- **Non-targeted-therapy comparison group:** the day after one year of continuous enrollment within a ≥2 year period between Jan 1st, 2016, and Dec 31st, 2023
- **Targeted therapy group:** first prescription of targeted therapy between Jan 1st, 2017, and Dec 31st, 2022

Follow-up period:

- **Pre-index period:** ≥12 months prior index date
- **Post-index period:** ≥12 months after index date

Washout period:

Targeted therapy group – not prescribed a targeted therapy in the year before the index date.

therapy, conventional systemic therapy, or supportive care, we refer to this group as the nontargeted-therapy comparison group. We did not further distinguish among patients in the comparison group based on alternative treatments because we were not able to identify any treatments received over the counter, and the current broad definition of comparison group reflects the real-world landscape of AD population who did not receive the targeted therapy, including both those untreated and those with alternative therapies, forming a meaningful comparison with our targeted-therapy group. For the comparison group, the index date was 12 months after the start of continuous enrollment period. For patients in the comparison group with multiple continuous enrollment periods, we kept the first instance only.

Patients were excluded if they did not meet continuous enrollment requirements, were aged younger than 18 years during the continuous enrollment period, or did not meet the claims-based AD definition.

STUDY OUTCOMES

The primary outcomes of interest were 2 indicators for health care use: (1) any all-cause hospitalizations and (2) any all-cause ED visits. We selected all-cause hospitalization and ED visits because they capture high-intensity acute care use relevant to patients, clinicians, and payers. These outcomes were intended to measure population-level, payer-relevant acute care use after targeted-therapy initiation rather than to attribute each encounter directly to AD or treatment. Thus, the estimates do not identify whether the specific encounter was directly attributable to AD, treatment failure, treatment-related adverse events, or other conditions. We did not restrict to AD-specific or adverse event–specific ICD-10 codes, because acute care driven by poorly controlled AD, such as secondary skin infections, is often coded to the proximate clinical diagnosis rather than to AD itself. A code-restricted definition therefore risks excluding clinically severe, policy-relevant encounters and understating true acute care burden. Any AD-unrelated encounters captured in these outcomes are addressed by our DiD design, which accounts for fixed differences in baseline levels between groups and would only introduce bias if trends in AD-unrelated encounters diverged systematically between groups over time, which there is no clinical reason to expect.

The secondary outcomes of interest were annual all-cause total health care spending and spending categories including drug, inpatient, and outpatient spending. Total health care spending was defined as the sum of inpatient, outpatient, and drug spending. Drug spending included all prescription drug spending captured in claims and was not limited to AD-specific medications. Out-of-pocket (OOP) spending was also examined, including total OOP spending as well as drug, outpatient, and inpatient OOP spending. All spending values were winsorized at the 99th percentile to mitigate the influence of outliers and adjusted for inflation using the medical care component of the consumer price index to reflect values in 2023 dollars.[16]

STATISTICAL ANALYSIS

We summarized the baseline patient characteristics using mean and SD for continuous variables and percentages for categorical variables. These included the patient's sex, age, geographic region, and targeted therapy prescribed (if any) on the index date. The Charlson Comorbidity Index (CCI) was calculated using comorbidity, an R package, which uses ICD-10 diagnosis codes on medical claims collected during the 12-month preindex period.[17–20] We assessed the differences in baseline characteristics between the 2 groups

using the Mann-Whitney test for age and chi-square test for sex, region, and CCI.

We employed a DiD study design to estimate the impact of targeted therapy on outcomes. The DiD design compares changes in outcomes before and after the index date between the targeted-therapy group and the nontargeted-therapy comparison group (see Supplementary Methods 1 for detailed model specification).

In the DiD model, the coefficient of interest was the interaction between the targeted-therapy group indicator and the postindex period indicator. The model adjusted for person fixed effects to control for time-invariant individual characteristics and calendar year-month fixed effects to control for secular trends, seasonality, and changes in health care practice common to both groups. We additionally controlled for time-varying comorbidity burden using CCI. We did not separately measure all possible time-varying clinical factors such as dermatologic severity, immunologic conditions, or use of nontargeted AD therapies. These factors were not included because they typically change in response to treatment itself, and adjusting for them would risk removing part of the true treatment effect rather than controlling for confounding. SEs were clustered at the individual patient level. In addition, spending outcomes were analyzed as continuous outcomes using linear DiD models to estimate adjusted mean differences in spending. We used linear models because mean spending differences are directly interpretable for payer and managed care decision-making and additionally estimated Poisson models for spending outcomes in sensitivity analysis.

The DiD method requires the parallel trends assumption to yield plausibly causal estimates of treatment effects. The assumption posits that, in the absence of targeted-therapy initiation, the targeted-therapy group and nontargeted-therapy group would have followed similar trends over time. We assessed the plausibility of the parallel trends assumption through visual inspection of the mean outcomes for both groups over time.

All statistical analyses were performed using R Statistical Software (v4.2.2; R Core Team, 2024).

Results

BASELINE CHARACTERISTICS

We identified a total of 157,966 individuals diagnosed with AD in the MarketScan database between January 1, 2017, and December 31, 2022, who met the eligibility criteria. These included 148,646 nontargeted-therapy comparison individuals and 9,320 individuals who received targeted therapy (Figure 2), 99% of whom were prescribed dupilumab (Table 1). The final analysis sample included 796,254 person-years, including 51,738 person-years from the treated group and 744,516 person-years from the comparison group. The distribution of person-years across all years for the treated and comparison group is reported in Supplementary Figure 1.

The mean [SD] age at baseline was 41.1 [13.6] years for the treated group and 41.6 [14.1] years for the comparison group ($P = 0.008$) (Table 1). The treated group had a higher proportion of men than the comparison group (42% vs 34%; $P < 0.001$). The treated group had more individuals from the South, whereas the comparison group had more from the Northeast and West ($P < 0.001$). The treated group had a lower proportion of individuals with a CCI score of 0 (46% vs 57%) and a higher proportion with a CCI score of at least 3 (13% vs 10%; $P < 0.001$).

DiD RESULTS ON PRIMARY OUTCOMES

The unadjusted trends in the probability of any hospitalizations (panel A) and the probability of any ED visits (panel B) are shown in Supplementary Figure 2. The trends in the probability of any ED (panel B) visits were generally parallel between the targeted-therapy group and nontargeted-therapy comparison group prior to the index date. The trends in the probability of any hospitalization show greater variability but are broadly consistent with the parallel trends assumption. Table 2 shows the unadjusted means by group and time period for the primary outcomes and the adjusted treatment estimates using DiD regressions.

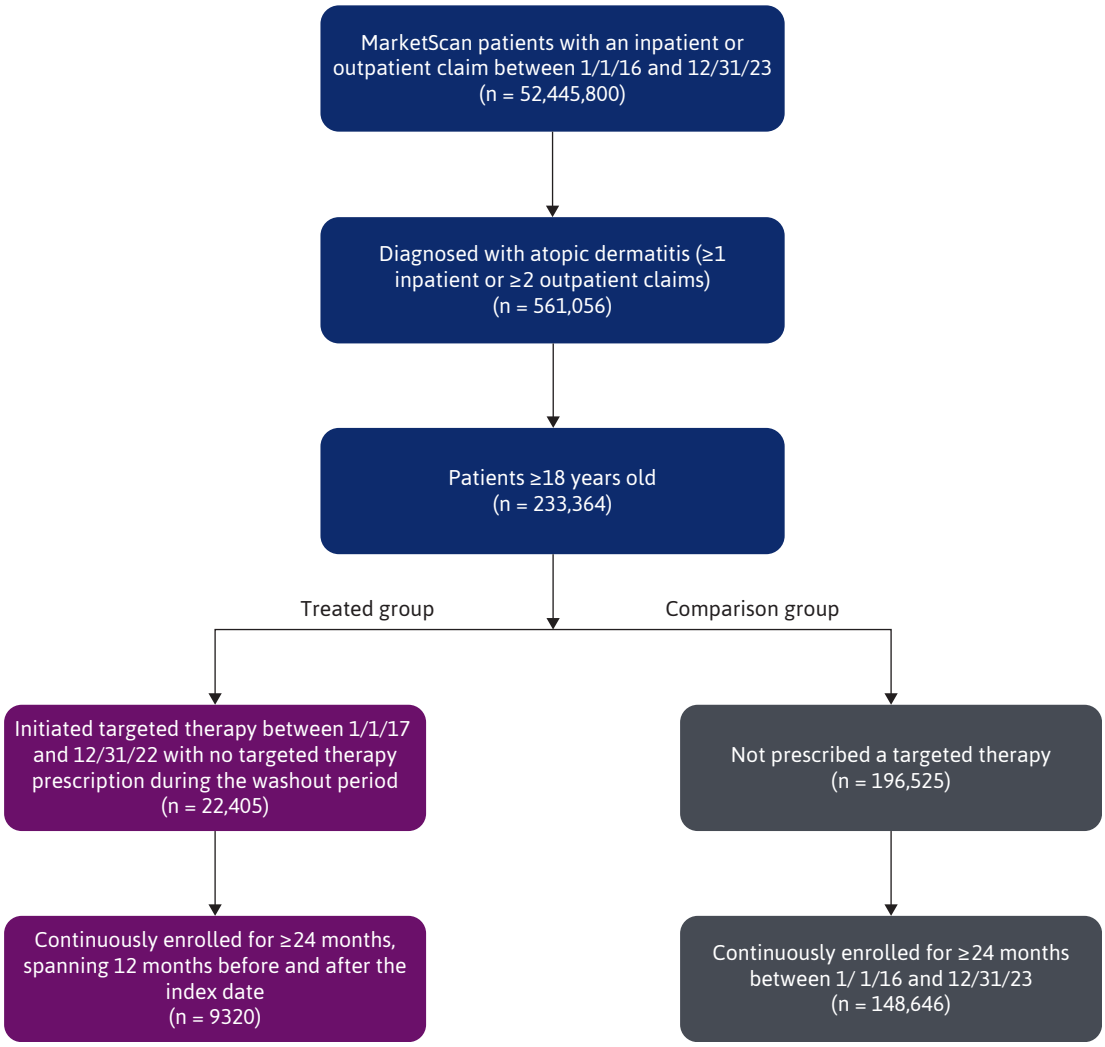
In unadjusted analyses, the proportion of subjects with any hospitalization decreased from the preindex to the postindex period in both the comparison group (from 7.0% to 6.8%) and the targeted-therapy group (from 7.2% to 6.0%). The proportion of patients with any ED visit increased slightly in the comparison group (from 16.2% to 16.9%) and decreased in the targeted-therapy group (from 19.0% to 17.9%) over the same period.

After controlling for covariates and fixed effects, targeted-therapy initiation was associated with an additional 1.6-percentage-point (95% CI=−0.021 to −0.010; $P < 0.01$) decrease in the probability of any hospitalizations and an additional 1.7-percentage-point (95% CI=−0.025 to −0.008; $P < 0.01$) decrease in the probability of any ED visits compared with the nontargeted-therapy comparison group.

SECONDARY OUTCOMES

The unadjusted trends for the health care spending outcomes are shown in Supplementary Figure 3. Preindex spending trends were generally similar for most of the preindex period. However, total health care and drug spending increased among targeted-therapy groups in the year immediately before the index date. Therefore, spending estimates should be interpreted more cautiously in our study context.

FIGURE 2 Cohort Selection and Attrition Diagram



The unadjusted means results for health care spending showed increases across all categories in the comparison group between the preindex and postindex periods, including mean total spending (from $7,979 to $8,955), drug spending (from $2,156 to $2,499), outpatient spending (from $4,827 to $5,396), and inpatient spending (from $996 to $1,060). In the treated group, total and drug spending rose sharply, with total spending increasing from $11,011 to $37,935 and drug spending from $4,216 to $31,000. Outpatient spending increased slightly (from $5,691 to $5,888), whereas inpatient spending saw a small decline (from $1,104 to $1,047) over the same period.

The adjusted results from the DiD regressions showed that targeted therapy was associated with a $24,901 (95% CI = $24,482-$25,320; $P < 0.01$) increase in annual total spending and a $26,061 (95% CI = $25,734-$26,387; $P < 0.01$) increase in annual drug spending compared with those not on a targeted therapy. Targeted therapy was also associated with a $909 (95% CI = −$1,106 to −$711; $P < 0.01$) decrease in outpatient spending and a $251 (95% CI = −$382 to −$119; $P < 0.01$) decrease in inpatient spending when compared with those not on a targeted therapy (Table 3). Results from Poisson regression specification (omitted) were consistent in direction and statistical significance with our primary linear DiD estimates.

Targeted therapy was also associated with a $95 increase in annual total OOP spending (95% CI = $83-$108; $P < 0.01$), driven by higher drug and inpatient OOP costs but lower

TABLE 1 Study Population Demographics and Comorbidities

Characteristics	Treated group (N = 9,320)	Comparison group (N = 148,646)	*P* value
Age, mean (SD), years	41.13 (14.14)	41.58 (13.57)	0.008
Sex, n (%)			
Female	5,408 (58.0)	97,481 (65.6)	<0.001
Male	3,912 (42.0)	51,165 (34.4)	
Region, n (%)			
Northeast	1,185 (12.7)	24,023 (16.2)	<0.001
North Central	1,232 (13.2)	20,012 (13.5)	
South	3,803 (40.8)	56,079 (37.7)	
West	959 (10.3)	19,008 (12.8)	
Unknown	2,141 (23.0)	29,524 (19.9)	
Comorbidity index, n (%)			
0	4,258 (45.7)	85,301 (57.4)	<0.001
1	3,027 (32.5)	34,951 (23.5)	
2	833 (8.9)	13,593 (9.1)	
≥3	1,202 (12.9)	14,801 (10.0)	
Targeted therapy, n (%)			
Abrocitinib	0	NA	
Baricitinib	8 (<0.1)	NA	NA
Dupilumab	9,247 (99.2)	NA	
Tralokinumab	1 (<0.1)	NA	
Upadacitinib	64 (0.7)	NA	
No treatment	NA	148,646 (100.0)	

NA = not applicable.

TABLE 2 Difference-in-Differences Results on Primary Outcomes

Outcome	Unadjusted means[a]				Adjusted treatment effect (95% CI)[b]
	Comparison group		Treated group		
	Preindex	Postindex	Preindex	Postindex	
Any hospitalization	0.070	0.068	0.072	0.060	−0.016*** (−0.021 to −0.010)
Any ED visit	0.162	0.169	0.190	0.179	−0.017*** (−0.025 to −0.008)

[a]*Unadjusted means are the raw means of 1-year outcomes for the given cohort in the period outlined.*

[b]*The adjusted treatment effect report results from linear regressions that control for person indicators, calendar year-month indicators, and Charlson Comorbidity Index score. Coefficients on the interactions between an indicator for the treatment group and an indicator for the postperiod are reported, with 95% CIs in parentheses. SEs are clustered at the individual patient level (*P < 0.1; **P < 0.05; ***P < 0.01).*

ED = emergency department.

outpatient OOP costs (Supplementary Table 3). Overall, OOP spending trends mirrored those of non-OOP spending, except inpatient spending decreased and inpatient OOP costs increased only slightly.

Discussion

We conducted a retrospective cohort study using MarketScan claims data to estimate the associations between targeted-therapy initiation, ED visits, hospitalizations, and health care spending among adults diagnosed with AD. On average, targeted therapy was associated with a 1.6-percentage-point decrease in the probability of any hospitalization and a 1.7-percentage-point decrease in the probability of any ED visits compared with the nontargeted-therapy comparison group. In relative terms, these estimates correspond to approximate 23% and 10% reductions, respectively, calculated by dividing DiD estimates by the corresponding comparison group rates (ie, 1.6 percentage points divided by 7.0% for hospitalization and 1.7 percentage points divided by 16.9% for ED visits).

Targeted therapies have been demonstrated to improve symptoms associated with AD such as skin clearance, reduction in itchiness, and decreased frequency of flares. The resulting improvement in symptom management can lead to a decreased likelihood of requiring an ED visit or hospitalization. Studies have also shown that targeted therapy can effectively manage AD symptoms, which can lead to a significant improvement in a patient's quality of life by enhancing sleep patterns and reducing anxiety and depression, perhaps further reducing the likelihood of related hospitalizations and ED visits.[21,22]

Targeted therapy was further associated with a $24,901 increase in mean total spending and $26,061 increase in mean drug spending while also contributing to a $909 decrease in mean outpatient spending and $251 decrease in mean inpatient spending. Despite a decrease in ED visits and hospitalizations, there was a significant rise in total spending, mainly driven by higher drug spending, which was likely

TABLE 3 Difference-in-Differences Results on Total Health Care Spending and Spending by Type

Outcome	Unadjusted means[a]				Adjusted treatment effect (95% CI)[b]
	Comparison group		Treated group		
	Preindex	Postindex	Preindex	Postindex	
Total spending, $	7,978.58	8,954.78	11,010.60	37,934.62	24,901.01*** (24,482.26 to 25,319.76)
Drug spending, $	2,155.83	2,498.88	4,215.51	31,000.12	26,061.26*** (25,734.560 to 26,387.960)
Outpatient spending, $	4,826.84	5,396.16	5,691.48	5,887.804	−909.25*** (−1,106.970 to −711.520)
Inpatient spending, $	995.91	1,059.75	1,103.61	1,046.69	−251.01*** (−382.273 to −119.739)

[a]*Unadjusted means are the raw means of 1-year outcomes for the given cohort in the period outlined.*

[b]*The adjusted treatment effect reports results from linear regressions that control for person indicators, calendar year-month indicators, and Charlson Comorbidity Index score. Coefficients on the interactions between an indicator for the treatment group and an indicator for the postperiod are reported, with 95% CIs in parentheses. SEs are clustered at the individual patient level (*P < 0.1; **P < 0.05; ***P < 0.01).*

due to the high prices of targeted therapies. Recent research also highlighted the substantial spending of targeted therapies, revealing that out of an average of $2,381 spent on all-cause health care spending for adults with AD, $1,997 (84%) was attributed to outpatient pharmacy claims, with 72% of that amount being spent on targeted therapies.[13] Compared with the control mean of the study period, targeted therapy was associated with a 280% increase in mean total spending and a 1,123% increase in mean drug spending. Meanwhile, the relative decreases in mean outpatient and inpatient spending were 17.8% and 24%, respectively. Although these results highlight the real-world tradeoff between reductions in intensive health care uses and increases in drug spending, the overall cost-effectiveness of targeted therapies remains uncertain. Cost-utility analyses that account for quality-of-life improvements have found targeted therapies such as dupilumab to be cost-effective compared with the standard of care, particularly for patients with more severe AD who experience greater improvement in quality of life.[23,24] This highlights that claims data alone cannot fully capture the comprehensive benefits of targeted therapy because they overlook factors such as indirect costs and other intangible benefits. Consistent with this interpretation, a cost-of-illness study demonstrated a substantial economic burden associated with AD, primarily driven by productivity loss from work absenteeism and diminished work performance.[25]

A similar trend was observed with the initiation of targeted therapy on OOP spending, which was associated with a $95 increase in mean total OOP spending (Supplementary Table 3). This increase was primarily driven by a $140 rise in mean drug OOP spending, partially offset by a $45 reduction in outpatient OOP spending. The disproportionate rise in drug spending relative to drug OOP spending suggests that cost-sharing mechanisms may help mitigate the financial burden of increased drug expenditures. The decline in both outpatient spending and outpatient OOP spending may indicate that targeted therapy is offsetting the need for outpatient services, potentially reflecting improved disease control. Additionally, we found a modest decrease in inpatient spending and virtually no change in inpatient OOP spending, likely due to the infrequent nature of inpatient visits in a low-acuity condition such as AD contributing to the noisy OOP spending estimates.

These findings have several implications for managed care pharmacy. Targeted therapy was associated with fewer all-cause hospitalization and any ED visits, suggesting that the effective systemic treatment for AD may reduce high-cost acute care use. Nevertheless, the modest declines in inpatient and outpatient spending were insufficient to offset the substantial rise in pharmaceutical spending, resulting in a net increase in total health care costs. Thus, managed care decision-makers should balance pharmacy budget impact with potential reductions in acute and urgent care use while also considering prior authorization criteria, patient affordability, and adherence.

To our knowledge, this is the first real-world study on the associations between targeted therapy, health care uses and spending among individuals diagnosed with AD. The percentage reductions in hospitalizations and ED visits were generally consistent with the reductions in outpatient and inpatient spending. These results may be used in future models that evaluate the real-world value of targeted therapy in treating patients with AD. Although our study reflects the current treatment landscape for moderate-to-severe AD, it is important to note that dupilumab accounted for most of the targeted therapy population. Key strengths of this study

include the use of a large commercial claims database, MarketScan, and a DiD design that controlled for time-invariant individual characteristics and calendar time shocks common to both groups.

LIMITATIONS

This study has several limitations. First, the population consisted of patients with AD with employer-sponsored commercial health insurance because we used MarketScan. Therefore, the findings may not be generalizable to patients with other types of insurance or uninsured populations. Second, because the comparison group included patients with AD who did not receive targeted therapy, these patients may still have received topical therapy, conventional systemic therapy, or supportive care. Thus, our estimates should be interpreted as the association of targeted-therapy initiation relative to nontargeted-therapy care.

Third, even though our models included a person fixed effect to control for time-invariant individual characteristics and a calendar year-month fixed effect to control for common secular trends and seasonality, residual unobserved confounding may remain. The validity of our estimates relies on the assumption that any unmeasured time-varying clinical factors trended similarly between the targeted therapy and comparison groups over the study period, and our estimates could be biased if these factors diverged systematically between groups.

Fourth, hospitalizations and ED visits were measured as all-cause outcomes, so individual encounters cannot be attributed with certainty to AD, treatment failure, treatment-related adverse events, or unrelated conditions. Our DiD design assumes that any AD-unrelated acute care use trended similarly between the targeted therapy and comparison groups over the study period; if this assumption does not hold, our estimates could still be biased.

Fifth, our 12-month outcome window did not require continuous treatment through the point of each encounter, so our estimates reflect an average effect across patients regardless of treatment persistence and may not represent the effect specific to patients who remained on targeted therapy throughout follow-up.

Sixth, claims data do not capture manufacturer copay coupons or other assistance programs, which may overstate patients' net OOP burden, particularly for targeted therapies in which such programs are common. This affects our OOP spending estimates only and not our total health care spending estimates, which capture the full cost of care regardless of payer.

Finally, most patients in the targeted-therapy group received dupilumab, so the findings may not be generalizable to newer targeted therapies or to individual biologic and JAK inhibitor agents.

Conclusions

In this retrospective cohort study of commercially insured adults with AD, initiation of targeted therapy was associated with lower probabilities of all-cause hospitalization and ED visits compared with nontargeted-therapy care. The use of targeted therapy was associated with lower inpatient and outpatient spending but higher total health care and drug spending, whereas patient OOP spending increased only slightly. These findings suggest that targeted therapy may reduce high-intensity acute care use among adults with AD but with important tradeoffs in pharmacy and total health care spending. Future studies should investigate further about AD-specific acute care use, treatment-related adverse events, and the longer-term value of targeted therapies across different patient populations.

DISCLOSURES

The authors declare no competing interests.

ACKNOWLEDGMENTS

This study was supported by the University of Washington Comparative Health Outcomes, Policy, and Economics (CHOICE) Institute AbbVie Fellowship program. AbbVie had no influence in the study design, execution, or decision to publish.

REFERENCES

1. Kolb L, Villarreal LA, Sathe NC. Atopic dermatitis. In: *StatPearls*. StatPearls Publishing; 2023. Accessed December 27, 2023. http://www.ncbi.nlm.nih.gov/books/NBK448071/

2. Ständer S. Atopic dermatitis. *N Engl J Med*. 2021;384(12):1136-1143. PubMed doi:10.1056/NEJMra2023911

3. Drucker AM, Wang AR, Li WQ, Sevetson E, Block JK, Qureshi AA. The burden of atopic dermatitis: summary of a report for the National Eczema Association. *J Invest Dermatol*. 2017;137(1):26-30. PubMed doi:10.1016/j.jid.2016.07.012

4. Kwa L, Silverberg JI. Financial burden of emergency department visits for atopic dermatitis in the United States. *J Am Acad Dermatol*. 2018;79(3):443-447. PubMed doi:10.1016/j.jaad.2018.05.025

5. Roekevisch E, Spuls PI, Kuester D, Limpens J, Schmitt J. Efficacy and safety of systemic treatments for moderate-to-severe atopic dermatitis: a systematic review. *J Allergy Clin Immunol*. 2014;133(2):429-438. PubMed doi:10.1016/j.jaci.2013.07.049

6. Stölzl D, Weidinger S, Drerup K. A new era has begun: Treatment of atopic dermatitis with biologics. *Allergol Select*. 2021;5(1):265-273. PubMed doi:10.5414/ALX02259E

REFERENCES *continued*

7. Reich K, Teixeira HD, de Bruin-Weller M, et al. Safety and efficacy of upadacitinib in combination with topical corticosteroids in adolescents and adults with moderate-to-severe atopic dermatitis (AD Up): results from a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet*. 2021;397(10290):2169-2181. PubMed doi:10.1016/S0140-6736(21)00589-4

8. Simpson EL, Bieber T, Guttman-Yassky E, et al; SOLO 1 and SOLO 2 Investigators. Two phase 3 trials of dupilumab versus placebo in atopic dermatitis. *N Engl J Med*. 2016;375(24):2335-2348. PubMed doi:10.1056/NEJMoa1610020

9. Eichenfield LF, DiBonaventura M, Xenakis J, et al. Costs and treatment patterns among patients with atopic dermatitis using advanced therapies in the United States: analysis of a retrospective claims database. *Dermatol Ther (Heidelb)*. 2020;10(4):791-806. PubMed doi:10.1007/s13555-020-00413-8

10. Schild M, Weber V, Thaçi D, et al. Treatment patterns and healthcare resource utilization among patients with atopic dermatitis: a retrospective cohort study using German health claims data. *Dermatol Ther (Heidelb)*. 2022;12(8):1925-1945. PubMed doi:10.1007/s13555-022-00773-3

11. Merative. Merative Marketscan research databases. 2023. Accessed July 18, 2024. https://www.merative.com/content/dam/merative/documents/brief/marketscan-research-databases-for-life-sciences-researchers.pdf

12. Fenske DC, Ding Y, Morrow P, et al. Comparing the burden of illness in patients with alopecia areata vs atopic dermatitis in the US population from a payer perspective. *J Manag Care Spec Pharm*. 2023;29(4):409-419. PubMed doi:10.18553/jmcp.2023.29.4.409

13. Manjelievskaia J, Boytsov N, Brouillette MA, et al. The direct and indirect costs of adult atopic dermatitis. *J Manag Care Spec Pharm*. 2021;27(10):1416-1425. PubMed doi:10.18553/jmcp.2021.27.10.1416

14. Kim RW, Lam M, Abuabara K, Simpson EL, Drucker AM. Targeted systemic therapies for adults with atopic dermatitis: selecting from biologics and JAK inhibitors. *Am J Clin Dermatol*. 2024;25(2):179-193. PubMed doi:10.1007/s40257-023-00837-w

15. Ray WA. Evaluating medication effects outside of clinical trials: new-user designs. *Am J Epidemiol*. 2003;158(9):915-920. PubMed doi:10.1093/aje/kwg231

16. Medical care in U.S. city average, all urban consumers, not seasonally adjusted. Published online July 22, 2024. https://data.bls.gov/toppicks?survey=cu

17. Quan H, Sundararajan V, Halfon P, et al. Coding algorithms for defining comorbidities in ICD-9-CM and ICD-10 administrative data. *Med Care*. 2005;43(11):1130-1139. PubMed doi:10.1097/01.mlr.0000182534.19832.83

18. Quan H, Li B, Couris CM, et al. Updating and validating the Charlson comorbidity index and score for risk adjustment in hospital discharge abstracts using data from 6 countries. *Am J Epidemiol*. 2011;173(6):676-682. PubMed doi:10.1093/aje/kwq433

19. Charlson M, Szatrowski TP, Peterson J, Gold J. Validation of a combined comorbidity index. *J Clin Epidemiol*. 1994;47(11):1245-1251. PubMed doi:10.1016/0895-4356(94)90129-5

20. Gasparini A. comorbidity: an R package for computing comorbidity scores. *J Open Source Softw*. 2018;3(23):648. doi:10.21105/joss.00648

21. Worm M, Francuzik W, Kraft M, Alexiou A. Modern therapies in atopic dermatitis: biologics and small molecule drugs. *J Dtsch Dermatol Ges*. 2020;18(10):1085-1092. PubMed doi:10.1111/ddg.14175

22. Labib A, Ju T, Yosipovitch G. Emerging treatments for itch in atopic dermatitis: a review. *J Am Acad Dermatol*. 2023;89(2):338-344. PubMed doi:10.1016/j.jaad.2023.04.057

23. Kuznik A, Bégo-Le-Bagousse G, Eckert L, et al. Economic evaluation of dupilumab for the treatment of moderate-to-severe atopic dermatitis in adults. *Dermatol Ther (Heidelb)*. 2017;7(4):493-505. PubMed doi:10.1007/s13555-017-0201-6

24. Zimmermann M, Rind D, Chapman R, Kumar V, Kahn S, Carlson J. Economic evaluation of dupilumab for moderate-to-severe atopic dermatitis: a cost-utility analysis. *J Drugs Dermatol*. 2018;17(7):750-756. PubMed

25. Ariëns LFM, van Nimwegen KJM, Shams M, et al. Economic burden of adult patients with moderate to severe atopic dermatitis indicated for systemic treatment. *Acta Derm Venereol*. 2019;99(9):762-768. PubMed doi:10.2340/00015555-3212